Gauge Theory Seminar: The Classical Bauer-Furuta Invariant

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1 Seiberg–Witten Theory

We start with a 4-manifold with a Spin^c-structure $\mathfrak{s}$. Recall that this is equivalent to the data of a Hermitian bundle S together with a Clifford multiplication $\rho : T^*X \to \mathrm{End}_{\mathbb{C}}(S)$. It turns out S splits into two Hermitian subbundles S^+ and S^-, on which $\rho(a) : S^{\pm} \to S^{\mp}$ induces isomorphism for any $a \neq 0 \in T^*X$.

In the classical Seiberg-Witten setup [Wit94,Mor96], we fix (or not fix) a reference Spin^c-connection A and write the Seiberg-Witten map as

$$\mu : (A + i\Omega^1(X;\mathbb{R})) \times \Gamma(S^+) \to \Omega^2_+(X;\mathbb{R}) \oplus \Gamma(S^-) \oplus \Omega^0(X;\mathbb{R})/\mathbb{R}$$
$$(A + ia, \phi) \mapsto (d^+a - q(\phi) + \frac{1}{2}F^+_{A^t}, D^+_A\phi + \rho(a)\phi, d^*a)$$

where d^+a is the self-dual part of da, $q(\phi)$ is the ρ^{-1} trace free part of $\phi\phi^* \in \Gamma(\mathrm{End}_{\mathbb{C}}(S))$, $F^+_{A^t}$ is the self-dual part of F_{A^t}, where A^t is the induced connection on $\det(S^+)$, D^+_A is the Dirac operator on positive spinors, and d^* is the codifferential.

The index of the Dirac operator is computed by the Atiyah–Singer index theorem [AS68]. Write $L = \det(S^+)$, so $c_1(L) = c_1(\mathfrak{s})$, and let $\sigma(X)$ be the signature.

Theorem 1.1 ([AS68]). *The complex index of $D^+_A : \Gamma(S^+) \to \Gamma(S^-)$ is*

$$\mathrm{ind}_{\mathbb{C}}\, D^+_A = \frac{c_1(L)^2 - \sigma(X)}{8},$$

where $c_1(L)^2$ means $\langle c_1(L)^2, [X]\rangle$. In particular $\mathrm{ind}_{\mathbb{R}}\, D^+_A = \frac{c_1(L)^2-\sigma(X)}{4}$, independent of A.

Sketch. Locally $S^+ = S^+_0 \otimes L^{1/2}$ for a genuine spinor bundle S^+_0, so the index theorem gives

$$\mathrm{ind}_{\mathbb{C}}\, D^+_A = \int_X \hat{A}(X)\, e^{c_1(L)/2} = \int_X \Big(\frac{c_1(L)^2}{8} - \frac{p_1(X)}{24}\Big),$$

and $p_1(X)[X] = 3\sigma(X)$ by the Hirzebruch signature theorem. Integrality is consistent with $c_1(L)$ being characteristic: $c_1(L)^2 \equiv \sigma(X) \pmod 8$. □

In the Bauer-Furuta paper [BF04], the equation they use is a globalized version of Seiberg-Witten map, where the base is the space of connections as follows:

$$\mathcal{A} := \big((A + \ker(d)) \times (\Omega^1(X;\mathbb{R}) \oplus \Gamma(S^+))\big)/\mathcal{G}_0$$
$$\mathcal{C} := \big((A + \ker(d)) \times (\Gamma(S^-) \oplus \Omega^2_+(X;\mathbb{R}) \oplus H^1(X;\mathbb{R}) \oplus \Omega^0(X;\mathbb{R})/\mathbb{R})\big)/\mathcal{G}_0$$

and the Seiberg-Witten map is

$$\mu : \mathcal{A} \to \mathcal{C}$$
$$(A, \phi, a) \mapsto (A, D_{A+a}\phi, F^+_{A+a} - q(\phi), a_{harm}, d^*a)$$

Claim 1.2. *$\mathcal{A}$ and $\mathcal{C}$ are S^1-equivariant Hilbert bundles over $Pic(X)$ and the Seiberg-Witten map μ is an S^1-equivariant, fiber-preserving map.*

Now $\mu = l + c$, where $l = D_A \oplus d^+ \oplus pr_{harm} \oplus d^*$ and $c : (\phi, a) \mapsto (0, F_A^+, 0, 0) + (a \cdot \phi, -q(\phi), 0, 0)$. (We write q rather than Bauer–Furuta's σ for the quadratic term, since $\sigma(X)$ denotes the signature.)

Claim 1.3. *c is compact.*

Claim 1.4. *The linear part l is Fredholm at any $A \in Pic(X)$, so $\mu = l + c$ is a compact perturbation of a family of linear Fredholm operators, and*

$$\operatorname{ind} l = \operatorname{ind} D - \underline{H^2_+(X;\mathbb{R})}$$

where the underline means the trivial bundle over $Pic(X)$.

Proof. 1. $\operatorname{ind} D_A$ has its own domain and codomain, so that's fine.
2. $d^+ \oplus d^*$ has kernel $\mathcal{H}^1(X)$, the harmonic group, and cokernel $\mathcal{H}^2_+(X)$.
3. The addition of this harmonic projection kills the kernel $\mathcal{H}^1(X)$ in the domain. □

Theorem 1.5 ([BF04, Thm. 3.1])**.** *μ satisfies the boundedness condition. In particular, it defines a stable cohomotopy class $[\mu] \in \pi^{b_2^+}_{S^1}(Pic(X); \operatorname{ind} D) = \mathbb{S}^{b_2^+(X)}_{S^1,H}(\mathrm{Th}(\operatorname{ind} D_X))$.*

The Bauer–Furuta invariant recovers the integer Seiberg–Witten invariant in the following way:

Theorem 1.6 ([BF04, Thm. 3.3])**.** *If $b_2^+ > b_1 + 1$, a homology orientation, i.e. an orientation of $H^1(X;\mathbb{R}) \oplus H^2_+(X;\mathbb{R})$, determines a map*

$$t : \pi^{b_2^+}_{S^1}(Pic(X); \operatorname{ind} D) \to \mathbb{Z}$$

Proof. A stable map can be represented by a pointed equivariant map

$$\mu : \mathrm{Th}(F) \to V^+$$

where $F \to Pic(X)$ is a finite-rank subbundle with $\operatorname{ind} D = F - Pic(X) \times V'$, $V = V' \oplus H^2_+(X;\mathbb{R})$, and V^+ is the one-point compactification.

Then we perturb such that $\mathrm{Th}(F)^{S^1}$ is sent to a subspace of $(V^+)^{S^1}$ of codimension $b_2^+ - b_1 \geqslant 2$. Then we can further perturb it such that a generic point in the complement of $(V^+)^{S^1}$ is a manifold M with free S^1-action on it. The homology orientation induces orientation on F and V, then we have a relative orientation between F and V, which will induce an orientation on M. Note that

$$\dim M = \operatorname{ind}_{\mathbb{R}} D + b_1 - b_2^+$$

why b_1? Because we are taking a family with base $Pic(X)$, which has dimension b_1. Plugging in the index of D,

$$\dim M = \frac{c_1(L)^2 - \sigma(X)}{4} + b_1 - b_2^+, \qquad \dim M/S^1 = \frac{c_1(L)^2 - 2\chi(X) - 3\sigma(X)}{4},$$

which is the usual expected dimension of the Seiberg–Witten moduli space (use $\chi = 2 - 2b_1 + b_2^+ + b_2^-$ and $\sigma = b_2^+ - b_2^-$).

Now take $\mathcal{L} = M \times_{S^1} \mathbb{C}$, this line bundle has first chern class in $H^2(M/S^1)$, so if $\dim M = 2k + 1$, we can just take

$$c_1(\mathcal{L})^k [M/S^1]$$

and this is an integer. □

2 Permutation of Manifolds

Theorem 2.1 ([Bau04, Thm. 1.1]). $[\mu_{X\#Y}] = [\mu_X] \wedge [\mu_Y]$.

Sanity Check: The smash product on the right is a map

$$\begin{aligned}\wedge : [\mathbb{S}^{\operatorname{ind} D_X}, \mathbb{S}^{b_2^+(X)}] \times [\mathbb{S}^{\operatorname{ind} D_Y}, \mathbb{S}^{b_2^+(Y)}] &\to [\mathbb{S}^{\operatorname{ind} D_X} \wedge \mathbb{S}^{\operatorname{ind} D_Y}, \mathbb{S}^{b_2^+(X)+b_2^+(Y)}] \\ &\cong [\mathbb{S}^{\operatorname{ind} D_X \boxplus \operatorname{ind} D_Y}, \mathbb{S}^{b_2^+(X)+b_2^+(Y)}] \\ &= [\mathbb{S}^{\operatorname{ind} D_{X\#Y}}, \mathbb{S}^{b_2^+(X\#Y)}]\end{aligned}$$

Here $\boxplus$ is the external direct sum over $Pic(X) \times Pic(Y) = Pic(X\#Y)$, using $\operatorname{Th}(E) \wedge \operatorname{Th}(E') = \operatorname{Th}(E \boxplus E')$. So implicitly, the authors showed that $\operatorname{ind} D_X \boxplus \operatorname{ind} D_Y \cong \operatorname{ind} D_{X\#Y}$.

Construction 2.2 ([Bau04]). Given n connected 4-manifolds, $\bigsqcup X_i$ is also a 4-manifold. Choose a separating neck $N_i \cong [-R, R] \times S^3$ in each connected component X_i. Then X_i splits into two parts, we call one the positive part and the negative part. We may permute the positive parts of X_i's by some permutation τ. This also induces an identification of $Pic(X) \to Pic(X^\tau)$ and equivariant bundle isomorphisms

$$\mathcal{A} \to \mathcal{A}^\tau$$
$$\mathcal{C} \to \mathcal{C}^\tau$$

More specifically, for any vector bundle $E \to X$, we fix an identification $E|_{\sqcup N_i} \cong \oplus_{i=1}^n F$. Then under the permutation, the vector bundles glued up along S^3's to give E^τ. We would like this permutation to give a map $\Psi : \Gamma(E) \to \Gamma(E^\tau)$.

The problem is a smooth section might not be glued up smoothly under the permutation. So we have to smooth it. Here's the way the authors did: given a section e, we can write $e = (e_1, ..., e_n)$, where e_i is the section on X_i. Then choose a bump function ψ that vanishes on $[-R, -1]$ and 1 on $[1, R]$, then choose a path $\gamma : [0, 1] \to SO(n)$ that starts at the identity and ends at the permutation matrix of τ. This needs $\tau \in A_n$; for odd τ (e.g. the transposition used below) replace the permutation matrix by a signed one in $SO(n)$, such as $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ for $n = 2$. The extra sign is multiplication by -1 on one summand, a bundle automorphism, so nothing below changes. Then we define

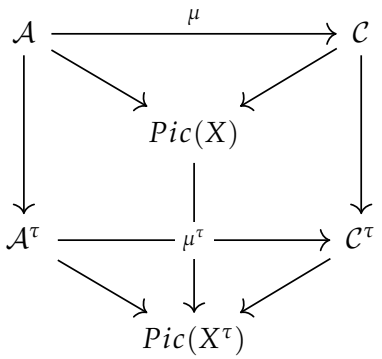
$$e^\tau = (\gamma \circ \psi)(e)$$

and we get a map $\Psi : \Gamma(E) \to \Gamma(E^\tau)$.

Claim 2.3. *The above map is an isomorphism of Hilbert spaces.*

The above maps are all constructed in this way.

Theorem 2.4 ([Bau04]). *The following diagram commutes up to a homotopy through Fredholm maps.*



Therefore, the induced cohomotopy map $[\mu]$ and $[\mu^\tau]$ are the same. So we have

$$\bigwedge_i [\mu_{X_i}] = [\mu_X] = [\mu_{X^\tau}] = \bigwedge_i [\mu_{X_i^\tau}],$$

where $X^\tau = \bigsqcup_i X_i^\tau$ is the decomposition into components (the class of a disjoint union is the smash product of the classes of its components).

Now we can form the disjoint union of $X \sqcup Y$, then by doing a transposition, we get $X\#Y \sqcup S^4$. The previous theorem tells us that

$$[\mu_X] \wedge [\mu_Y] = [\mu_{X\#Y}] \wedge [\mu_{S^4}]$$

Theorem 2.5. $[\mu_{S^4}] = [\mathrm{id}]$.

So we have the promised theorem.

References

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